
OOMWOO One

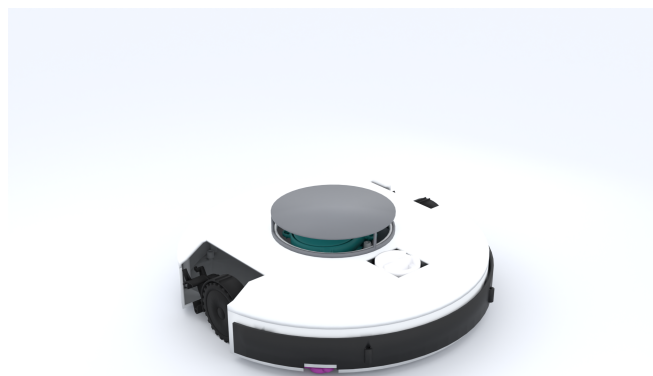
Full Closed Consumer Device — RV02 Report

From agentic-pipeline proof-of-concept to a sealed robot-vacuum: expanded component registry, a fully enclosing cover + turret cap, a functional-ready front bumper, and true bottom-extraction — every full-device gate verified on the built solids.

Prepared for	Ilia Ovsiannikov — makerspet / OOMWOO (makerspet/oomwoo-one-cad, Apache-2.0)
Engagement	Upwork Milestone 2 — “Detailed 3D modeling for the chassis”
Scope of RV02	Client-directed expansion of the POC into a complete <i>closed</i> device (§8)
Companion	OOMW00_Chassis_POC_Report.pdf (the pipeline proof-of-concept; unchanged)
Toolchain	build123d 0.10.0 / OpenCascade (OCP) booleans; multi-agent design + QA loop
Status	Phase 5B APPROVED — all full-device gates pass
Date	2026-07-25

What RV02 adds — and what remains a placeholder.

The proof-of-concept proved the *pipeline*: import real component STEPs, pack them in one coordinate frame, auto-generate a printable chassis, gate it numerically, and hot-swap a component with a one-line change. RV02, at the client’s direction, closes that open chassis into a coherent consumer product — a low-profile domed cover with a removable LiDAR turret cap, a functional-ready front bumper, a side-brush pair, a UI button cluster, a custom I/O board and a custom bottom-extraction dustbin. The functional components are still dimensioned *placeholders* (repo LiDAR/CM4 + the reverse-engineered A8S library) awaiting 3D scans; the swap protocol means each scan drops in as a one-line registry edit. All numbers below were re-measured on the built solids by an independent QA agent; the designer agent’s claims were not trusted.



The closed RV02 device — domed cover, removable LiDAR turret cap, recessed UI button, wrap-around front bumper, exposed drive wheel and front-corner side brush.

1. Executive summary

RV02 takes the approved proof-of-concept chassis and, per `DESIGN_SPEC.md §8`, expands it into a complete closed consumer robot vacuum. The ten POC components did not move; five new functional elements were added, the base was amended for true bottom-extraction and for the new enclosure interfaces, and a three-part outer shell (cover, turret cap, bumper) was designed and verified around the whole assembly. The device now reads as a finished product and passes every acceptance gate in §8.5.

- (a) **Expanded registry (10 → 15 components):** added a side-brush + its motor, a UI button cluster, a custom I/O board and a custom thin-wall dustbin — all placed without interference against the existing ten.
- (b) **Fan relocation for true bottom extraction (client decision):** the suction fan was moved rear-left so the dustbin can seat on the base floor and route a Ø52 mm extraction port through the floor within the mop-exempt bin footprint.
- (c) **Enclosure:** a full-Ø355 low-profile domed cover (parting line at the base rim, $Z=60$), a removable LiDAR turret cap with a 360°-clear sensor window, and a functional-ready 150° front bumper with travel, spring seats and guide followers.
- (d) **Full-device verification:** cover clears all 15 components (0 mm^3), seats on the base at a $\sim 0.04\text{ mm}$ parting gap, the bumper sweeps clean to a hard-stop, and the closed device stands 108 mm tall inside the height budget.

Headline result.

The closed device is verified. $\text{Cover} \cap \text{every one of the 15 components} = 0$, $\text{cover} \cap \text{base} = 0$ with an $\sim 0.04\text{ mm}$ parting gap, turret cap slip-fits at 0.30 mm over a 360°-clear sensor window, the front bumper travels its full stroke to an intended tangent hard-stop with zero clash to cover or side-brush sweep, and the recessed UI button sits $\sim 1.5\text{ mm}$ proud. Total device height is **108 mm** (≤ 115 budget); device diameter is Ø355 mm bare, Ø367 mm across the bumper. Every printed part has a stated print orientation and split. **QA verdict: Phase 5B APPROVED, no blocker or major findings.**

Closed device at a glance

Parameter	Basis	Value
Device height (incl. turret cap)	Measured on the assembled solid; budget ≤ 115	108.00 mm
Device diameter	Bare cover / across front bumper	Ø355 / Ø367 mm
Base (chassis) volume	Final, after side-brush motor-boss fix	481,399 mm ³
Cover / bumper / cap volume	Three printed shell parts	228,018 / 51,607 / 44,754 mm ³
Cover $\cap$ 15 components	Boolean common volume, all parts	0 mm ³
Parting-line gap (cover↔base)	Re-measured at 4 angles, $Z=60$	$\sim 0.04\text{ mm}$
Bumper travel (rest → hard-stop)	150° arc, sweep-verified	2.86 mm → 0
Dustbin capacity (thin-wall, 1.2 mm)	Custom bottom-extraction bin	$\approx 0.189\text{ L}$
Fasteners (heat-set / screw / align)	Full base + enclosure inventory	48 features

2. What's new in RV02

2.1 Expanded component registry (10 → 15)

The registry-driven pipeline made the expansion additive: five new entries were placed and gated against the existing ten with a single 15-solid interference pass (**0 clashes**). The custom parts (I/O board, dustbin) are parametric — their single source of truth is `custom_parts.py`.

New component	Source	Placement / key numbers (measured)
side_brush	A8S-U04 (placeholder)	Front-right corner (116, 116); swept radius 20.84mm, floor clearance 1.50 mm, edge overhang 7.39 mm (≤ 10)
side_brush_motor	A8S-M06 (placeholder)	Inboard at (105, 78) driving the corner disc via an implied arm / Ø8 pass-through (flagged — see §7)
ui_button	A8S-E04-01 (placeholder)	Front-center (0, 105); lowered to sit ~ 1.5 mm proud of the cover; cutout from measured bbox +0.4 mm
io_board	custom parametric plate	54.2×39.2×1.6 mm on the compute deck, connector edge $-Y$ (rear-accessible)
dustbin	custom parametric bin	57×56×66 mm thin-wall (1.2 mm), $\approx 0.189 L$, Ø52 bottom port, removable from top

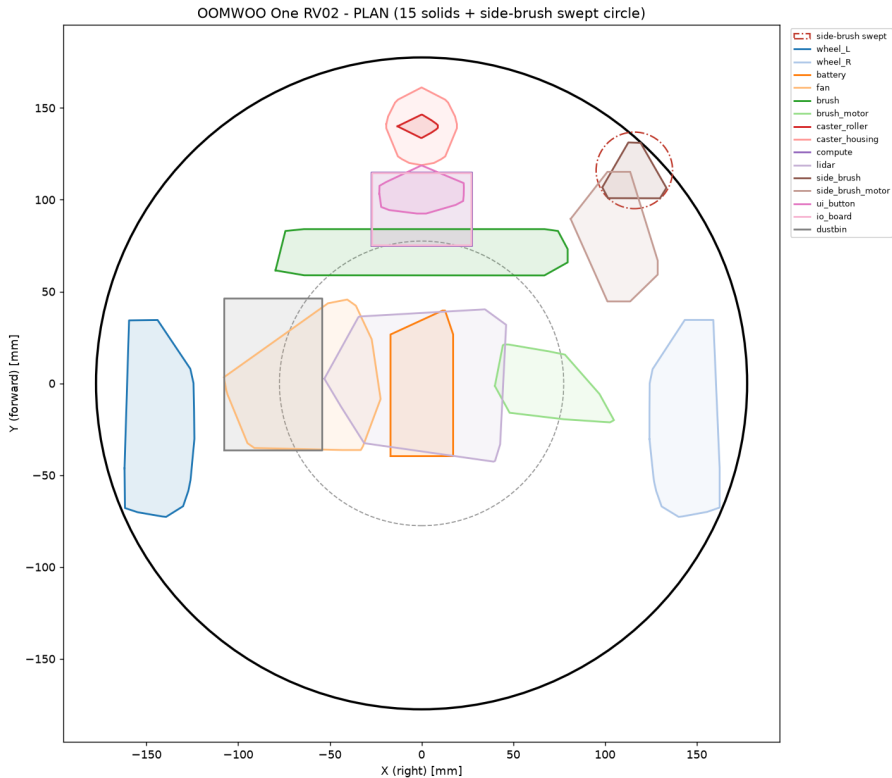


Figure 1. RV02 expanded plan — all 15 solids packed inside the Ø355 mm envelope, including the new front-right side brush (with its dash-dot swept circle), the front-center UI button, the custom I/O board and dustbin. The ten POC components are unchanged.

2.2 Fan relocation for true bottom extraction (client decision)

In the POC the suction fan and dust-bin bay were a single *reserved open volume* at front-left. To realize the spec’s bagless, bottom-extraction auto-empty (§2/§8.1), the client directed that the fan be relocated so the bin can reach the base floor. The fan was moved **rear-left** (registry position (−65, −52); measured bbox Y[−93.3, −10.7]); the fan-bay pocket, being derived from the measured bbox, tracked the move automatically. The dustbin then dropped onto the base floor (Z[12, 78]) directly above a **Ø52 mm extraction port** cut through the floor, ringed by a 1 mm-recessed, 4 mm-wide sealing land. Port-to-bin concentricity and bin floor-seat both measure 0.00 mm; the dustbin-to-fan gap after the move is 1.20 mm.

2.3 Base amendments (Phase 5A)

Every previously approved feature and placement was retained; the base gained only additive features, and one bugfix took its final volume to **481,399 mm³**:

- **Side-brush motor-boss bugfix** (`_SBM[3]`→`_SBM[4]`) — corrected the boss index; this is the source of

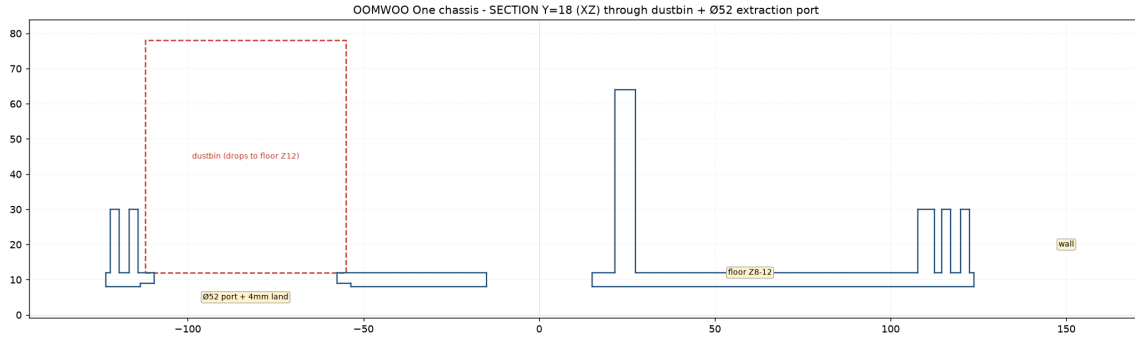


Figure 2. Base amendment — XZ section at $Y=18$ through the dustbin and the $\text{Ø}52\text{ mm}$ bottom extraction port. The thin-wall bin drops to the floor ($Z=12$); the port pierces the base floor ($Z=8-12$) with a recessed sealing land. Framed with margin, no clipping.

the base-volume change from the POC baseline.

- **Side-brush recess** $\text{Ø}43.7$ at (116, 116), edge-open, with 1.00 mm disc-to-pocket clearance; motor gets 2 M3 bosses at (76, 44) / (76, 112) plus a $\text{Ø}8$ drive-arm pass-through.
- **Bottom-extraction port + sealing land** (above) within the mop-exempt bin footprint.
- **Bumper interface** on the front inner wall — 2 spring bosses ($\pm 86, 149$), 2 guide slots ($\pm 155.9, 72.7$), and 4 front-wall pass-through windows for the bumper followers.
- **Cover interface** — 8 M3 cover-mount bosses (screw-from-below through-holes) + 2 alignment blind-holes.

3. Enclosure: cover, turret cap & bumper

3.1 Top cover

A single full- $\text{Ø}355\text{ mm}$ low-profile domed shell closes the device, with the parting line at the base rim ($Z=60$) and a smooth consumer top that exposes nothing but the LiDAR turret and the UI button. Measured shell walls are 2.45 mm (skirt and dome) and 3.00 mm at the turret collar — all $\geq 2.0\text{ mm}$. The cover fastens to the base with **8 M3 heat-set bosses, screwed from below** through the base, and locates on **2 alignment pins** seating in base blind-holes; it also carries the LiDAR turret opening + collar, the UI-button cutout with two support brackets, and two wheel-well arch cutouts.

3.2 LiDAR turret + removable cap

The turret collar (inner/outer radius 64.5 / 67.5 mm, top $Z=87.0$) rises from the measured LiDAR footprint, and a removable printed cap slips over it with a **0.30 mm** fit (cap inner radius 67.80 vs collar 67.50 — within the 0.2–0.4 mm target). The cap roof sits at $Z=105.5-108.0$; the sensor window band ($Z=88.5-102.5$) is **360° clear** — no cover material in the annulus, crossed only by the three standard $\text{Ø}3.5\text{ mm}$ cap-support legs. All of this regenerates from the measured LiDAR on a scan swap.

3.3 Front bumper

The bumper is a printable 150° front-arc shell (inner/outer radius 180.5 / 183.5 mm) carrying two spring seats ($\pm 30^\circ$), two guide followers ($\pm 65^\circ$) that pass through the four base wall windows, and a side-brush relief scallop. It rests **2.86 mm** off the base (spec target 2–3 mm) and travels rearward to an intended tangent hard-stop against the front wall at -3.0 mm , with the button 1.5 mm proud unaffected. Springs and detents are not modeled (bosses/followers only), consistent with §8.3.

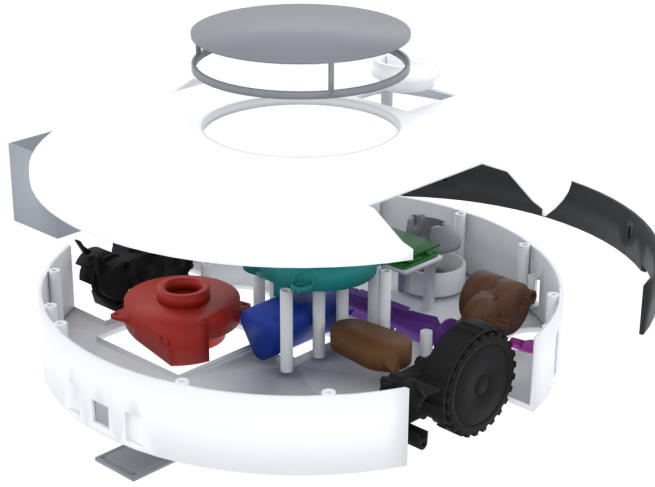


Figure 3. Exploded view — removable turret cap (top), full-Ø355 domed cover lifting off the base at the $Z=60$ parting line, and the front bumper separating forward. All internal components remain seated on the base.

The button is consumer-flush.

In Phase 5B the UI button was lowered (registry $Z\ 98.3 \rightarrow 75.3$) so its measured top (85.00 mm) sits **1.52 mm proud** of the cover outer surface (83.48 mm) at (0, 105) — a normal button feel. Its cutout ring holds a 0.40 mm gap (bbox +0.4/side) and it clears the nearest part below (the I/O board) by 2.41 mm. The cutout and its two bracket bosses regenerated from the newly placed bbox — measure-driven, not hand-tuned.

4. Full-device verification

All values re-measured independently on the built solids (build123d 0.10.0; OCP booleans, SolidClassifier, DistShapeShape). The compute module remains a pinned proxy box (CM4 not imported).

Gate (§8.5)	Measured	Requirement	
Part validity (cover / bumper / cap / base)	all valid, 1 solid each	valid solids	PASS
Cover wall (skirt / dome / collar)	2.45 / 2.45 / 3.00 mm	≥ 2.0 mm	PASS
Cover $\cap$ each of 15 components	0.0000 mm ³ on all 15	0	PASS
Cover $\cap$ base (intended contacts only)	0.0000 mm ³ ; 11–12 contact patches	0 material	PASS
Parting-line gap (4 angles, $Z=60$)	~ 0.04 mm	≤ 0.5 mm	PASS
Per-component cover headroom	≥ 2.50 mm (tightest: dustbin)	≥ 0.4 mm	PASS
Turret cap slip fit	0.30 mm	0.2–0.4 mm	PASS
LiDAR sensor window 360° clear	0 mm ³ in band (3 cap legs only)	clear	PASS
Bumper rest gap $\rightarrow$ full-travel	2.86 $\rightarrow$ 0.00 mm (tangent hard-stop)	2–3 mm rest	PASS
Bumper $\cap$ base / cover / side-brush sweep	0.0000 mm ³ through full travel	no clash	PASS
UI button proud height / clearance below	1.52 mm proud / 2.41 mm to I/O board	flush / ≥ 0.4	PASS
Dustbin fit / port concentricity	bay gap ≥ 1.20 mm / 0.00 mm	≥ 0.4 / aligned	PASS
Device height / max plan radius	108.00 mm / 183.50 mm	≤ 115 / —	PASS

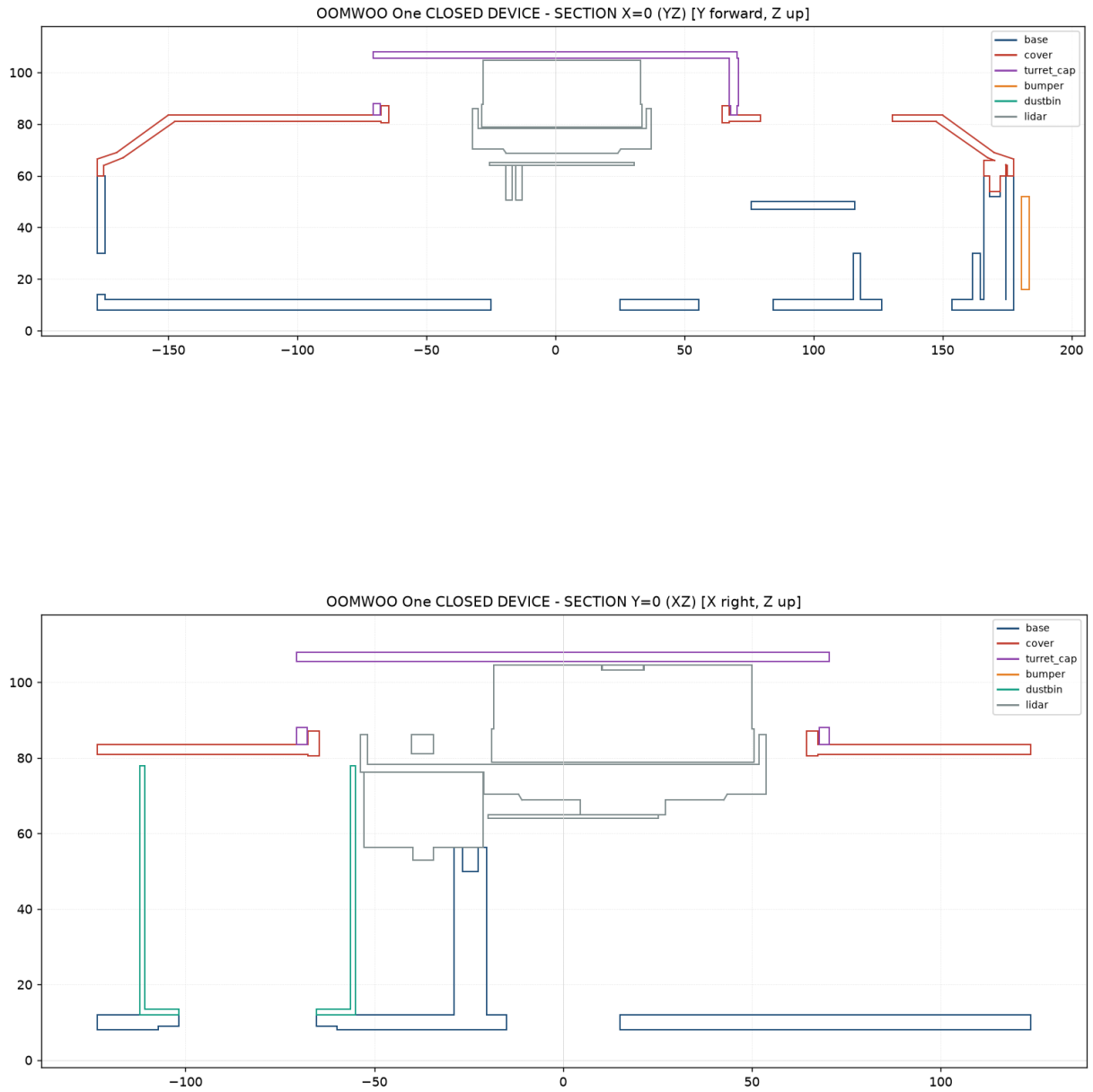


Figure 4. Closed-device cross-sections. *Top:* $X=0$ (YZ). *Bottom:* $Y=0$ (XZ). Both show the base, the domed cover meeting it at the $Z=60$ parting line, the removable turret cap over the LiDAR, the dropped dustbin, and the front bumper — a coherent sealed robot vacuum, framed with margin.

5. Print & assembly

5.1 Per-part print orientation & split

Every printed part carries a stated orientation and, where it exceeds the print bed, a split strategy. The cover and base exceed Ø350 and must be split; the cap, dustbin and battery door print in one piece.

Part	Size (mm)	1-pc fit?	Orientation / split
base	348×355×64	no	Floor-down; Ø355>350 → quadrant split at $X0\&Y0$ (4 D-parts $\approx 180\times 180$), dowel + M3 lap joints
cover	348×355×33	no	Dome-up; Ø355>350 → half split at $X=0$ (2 parts $\approx 355\times 178$), glue/screw rim seam
turret_cap	141×141×24	yes	Roof-up, legs down; fits 256 ³
bumper	354×137×36	no	Arc; split at Y-axis into L/R halves (or print on its back); fits after split
dustbin	57×56×66	yes	Open-top-up; walls 1.2 / floor 1.5 / ribs 2.2 mm
battery_door	38×58×2	yes	Print flat

5.2 Suggested assembly order

1. Print the base quadrants, install heat-set inserts in all bosses, and bond the quadrants (dowels + M3 lap joints) into the full base tub.
2. Seat the functional components on the base: drive wheels, caster, main brush + motor, fan (rear-left), battery, compute + I/O board, LiDAR, side brush + its inboard motor.
3. Drop the **dustbin** into its floor bay from the top (before the cover) so it registers over the extraction port; fit the **battery door** plate from underside.
4. Lower the **cover** onto the base, locating on the 2 alignment pins, and fasten the **8 M3 screws from below** through the base into the cover bosses.
5. Press the **turret cap** onto the collar (0.30 mm slip fit) over the LiDAR.
6. Fit the **bumper**: engage its guide followers through the four front-wall windows onto the spring bosses (springs/detents added at the functional-prototype stage).

5.3 Fastener / mount inventory (48 features)

Feature	Count	Feature	Count
LiDAR seat cradle posts (M3)	8	Battery-door bosses	4
Compute M3 holes	4	Door-plate pads	4
Brush bearing bosses	2	Side-brush motor bosses	2
Brush-motor bosses	2	Bumper spring bosses	2
Wheel-module mounts	4	Bumper guide slots	2
Cliff-sensor bosses	4	Cover-mount M3 bosses	8
		Cover alignment pins	2
		Total heat-set / screw / alignment features	48

6. Assembly views



Figure 5. *Left:* closed top — domed cover, gray LiDAR turret cap, recessed front UI button, and the wrap-around front bumper with its two wheel-well arch cutouts. *Right:* underside — relocated fan (red), I/O board plate, main brush (purple), 3-arm side brush (magenta), caster ball, drive wheels and the bottom extraction port.

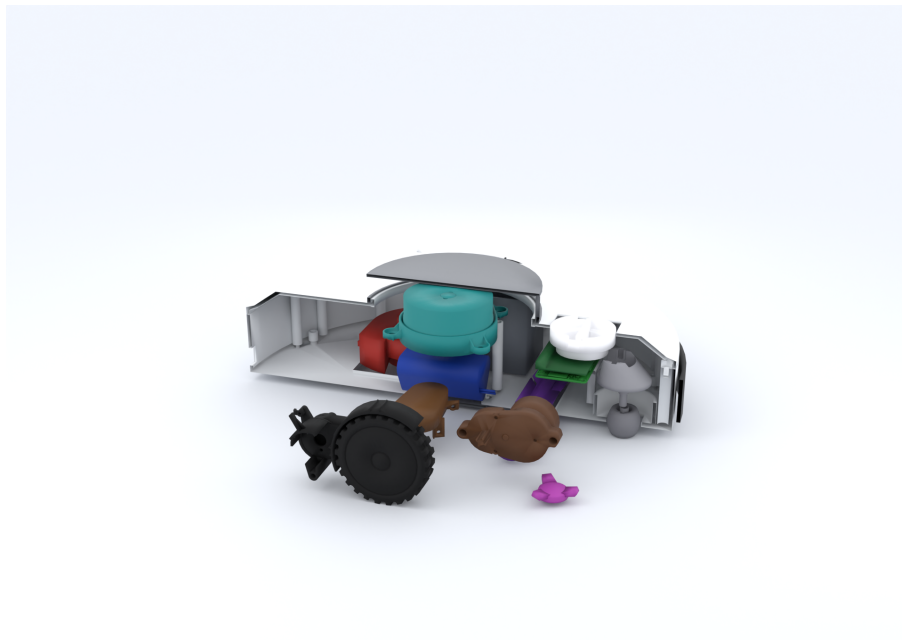


Figure 6. Cutaway — the sealed device with the cover and turret cap in place, showing the internal packing: LiDAR on its cradle, fan, dustbin, battery, drive train and side-brush corner unit all enclosed beneath the domed top.

7. Honest limitations & next steps

RV02 is a complete closed-device model built on placeholder internals; the following are the known limitations and the natural next steps once the 3D scanner arrives.

1. **Placeholders pending scans.** All functional components remain dimensioned placeholders (repo LiDAR/CM4 + the reverse-engineered A8S library). The turret collar/cap/window, seat cradle and every pocket derive from the *measured* placeholder and regenerate on a one-line registry swap — so each scan drops in without a redesign. Fit accuracy improves when the scans land.
2. **Bumper springs / detents unmodeled.** The bumper carries follower geometry and wall pass-throughs only; the return springs and any retention detent are added at the functional-prototype stage (per §8.3). Full travel is an intended tangent hard-stop on the front wall.
3. **Side-brush drive is implied.** The placeholder side-brush motor is too large to sit directly above the corner disc without clashing the main brush or exceeding Ø355; it is placed inboard and drives the disc through an implied arm/gearbox (Ø8 pass-through). A real gear-motor selection would firm this up.
4. **Dustbin volume is modest ($\approx 0.189\text{ L}$).** The thin-wall bin fits the available floor pocket beside the relocated fan; capacity is limited by the packing. Enlarging it (or trading against runtime between auto-empty cycles) is a layout decision for the next pass.
5. **Volute is a later CFD task.** The fan bay is a reserved volume; the detailed suction volute/duct is intentionally not designed here and is a downstream, CFD-driven task.
6. **CM4 compute is a proxy.** The compute module is a pinned proxy box; the real CM4/CM5 STEP swaps in as a one-line registry change when its mount references are needed.

Assumptions to confirm with the client / A8S owner

These carry over from the specification's open-questions log and gate the transition from placeholders to production intent:

1. **A8S → function mapping** — the reverse-engineered A8S part identities were proposed by geometry inspection and reconciled against the A8S BOM; please confirm against the physical parts.
2. **Overall diameter & height** — Ø355 mm / 108 mm is a designer recommendation (diameter is a spec design variable); confirm against your intent. Height is LiDAR-sensitive and drops with a thinner scanned sensor.
3. **Compute** — CM4 assumed (same footprint as CM5 per the repo); confirm CM4 vs CM5.
4. **I/O board outline** — modeled as a parametric rectangle with a rear-accessible connector edge per the spec's "decided by the CAD"; confirm the outline and connector side.
5. **Dustbin strategy** — $\approx 0.189\text{ L}$, top-removable, bottom-extraction port; confirm capacity vs. auto-empty cadence and dock-port geometry.

Recommended next steps

- **Scan swap-ins** — replace each placeholder via its registry line as scans arrive; re-run the fit loop per component and re-gate.
- **Dock interface** — design the rear charging-contact + homing geometry and the auto-empty dock mating to the bottom-extraction port.
- **DFM pass** — draft angles, split-seam alignment/fastening detail, heat-set boss finalization, and a mold-ability review ahead of tooling.

All numeric values re-measured on the final build123d solids by an independent QA agent (build123d 0.10.0, OCP booleans); the designer agent's asserted numbers were not trusted. Companion to OOMWOO_Chassis_POC_Report.pdf. Prepared for Ilia Ovsianikov, Upwork Milestone 2, makerspet/oomwoo-one — 2026-07-25.